

A Note Comparing Three Incentive Designs Against Privacy-Targeted Collusion

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Abstract. We compare three recent works on collusion deterrence mechanisms [SP'24, Eurocrypt'25, CCS'25] in privacy-preserving multi-party computations. They follow the same whistleblowing structure where an evidence collection module collects collusion evidence, and a mechanism assigns payments to deter incentive-driven parties from collusion.

For evidence collection module, two works [SP'24, Eurocrypt'25] provide a general method for generating collusion evidence while tolerating pre-existing leakage. The other work [CCS'25] abstracts evidence generation away, except for transparent service applications where the output is treated as the evidence. For the incentive mechanisms, two works [SP'24, Eurocrypt'25] consider a mix of *rational* and *malicious* parties, and rational parties can act as an individual or as a member of a *strong coalition*, inside which parties trust each other and never harm other members. When parties act as individuals, given bounded malicious parties, one can design mechanisms to disincentivize collusion. When parties act as a coalition, the mechanisms can only limit the size of coalitions for exclusive secrets, i.e., more parties learning the secret reduces the value received by individuals. The most recent work [CCS'25] only models *rational* parties but considers colluding parties establishing retaliatory contracts to discourage betrayal among colluders. It was shown to be impossible to maintain non-collusion outcome if retaliatory contracts can impose unbounded penalties, and feasible to guarantee non-collusion otherwise. This is weaker than a strong coalition but admits mechanisms protecting secrets of a general nature.

1 Introduction

In general privacy-preserving computation tasks involving multiple parties such as information theoretic secure multi-party computation [3, 4, 18] (MPC) and multi-server private information retrieval [2, 5, 8] (PIR), privacy cannot be ensured if a subset of computing parties collude. Even for additive secret sharing style tasks such as dishonest majority MPC [1, 6, 9], where all parties need to collude to break privacy, privacy cannot be guaranteed for a client who outsources the computation.

Bit guessing game. Drawing intuitions from a secret-shared secret bit guessing game, Gong et al. [10] initiated the study of collusion deterrence mechanisms targeting privacy-preserving multi-party computation in 2022. We briefly recall the game description below to build intuition. Let there be an honest game host and k *rational* players who are incentive-driven and take actions that maximize their utilities.

1. The host commits to a secret unbiased bit b and additive-secret-shares it among k players so that b can be reconstructed with all k shares together.
2. After sharing, each player can make *open* guesses about b , and the game terminates after a certain time.
3. After termination, the host penalizes a player \$4 for a wrong guess. If any party guesses correctly, the *first* correct guesser is selected as the winner. The host rewards the winner \$2 but penalizes each of the other $(k - 1)$ players \$4.

In the guessing game, without reconstructing bit b , a player does not guess because the expected return is negative (\$-1). For players without a network advantage in submitting guesses first, not

collaborating to reconstruct b is strictly dominating (i.e., generating strictly more returns) since their expected return $(\frac{k-1}{k}(-4) + \frac{1}{k} \cdot 2) < 0$ for $k \geq 2$.

This simple thought experiment implies that we can make *unobserved* collusion visible and undesired for rational participants via a proper incentive structure. Gong et al. [10, Section 1.1] utilized this intuition and designed a collusion deterrence protocol (executed on top of the base primitive) with two parts: (Part 1) an evidence collection module receives and verifies collusion evidence; (Part 2) an incentive mechanism disincentivizes rational parties from collusion. Of course, their protocol is built under more realistic models than the bit guessing game, e.g., a biased secret, a mix of rational and malicious parties who behave arbitrarily, dishonest host, and strong coalitions where members do not report collusion. Later works [12, 14] followed the same intrinsic rationale to deter privacy-targeted collusion: colluding parties submit reports to whistleblow on collusion; with enough rational parties, non-collusion or not collaborating to reconstruct secrets becomes the equilibrium strategy.

Goal of this note. We aim to fairly summarize and compare the three works to give a better picture on how incentive mechanisms can be utilized to discourage rational parties from collusion. Given that such incentive mechanisms are relatively under-explored in privacy-preserving multi-party computations, we also describe how to interpret the results and outline open problems in the context of MPC in this note.

1.1 Comparison Summary

We compare the three works in detail below, which is summarized in Table 1 (evidence collection methods) and Table 2 (the performance and modeling of the three mechanisms). Note that the evidence collection routines are protocol-dependent, but all the mechanisms are agnostic to protocol details and generally applicable to privacy-preserving multi-party computations.

Evidence collection. Gong et al. [10, 12] provided a general recipe for proving collusion without making assumptions on a specific collusion method:

- (Step 1) One or more colluding parties report non-trivial¹ knowledge of a secret after successful collusion along with their inputs that contribute to this knowledge;
- (Step 2) Other implicated parties submit a (collaborative) zero-knowledge non-interactive succinct argument of knowledge [17] (zk-SNARKs) of inequality if collusion did not happen, or a proof of triviality [19] if the function f is trivial (up to a parameter).

Silence of the implicated parties signal collusion. Of course, as noted by the authors, one needs to factor in pre-existing knowledge, set up a reasonable evidence collection time window for (Step 2) to account for denial-of-service attacks, and potentially adopt a publicly auditable version of the collaborative zk-SNARKs protocol to accommodate malicious implicated parties being silent to sabotage the proof generation.

Kelkar et al. [14] focused on the incentive mechanism designs and abstracted away general evidence generation. The authors provided an approach to generate collusion evidence in transparent service applications where an unlogged output itself serves as the evidence.

The major difference between the two methods is whether a party can generate collusion evidence alone. Gong et al. [10, 12] considered general cases and arbitrary collusion methods, and a colluding

¹ Informally, knowledge $f(x)$ on a secret x is non-trivial if it is not easily guessable, which can be connected to and measured with min-entropy of a function on a random variable with existing bounds by Sason [19].

Table 1: Comparison of evidence collection methods in two privacy-targeted collusion evidence collection methods (in chronological order).

Works	Methods ²
[10, 12]	(i) A strict subset of colluding parties report knowledge gain; (ii) Rest implicated parties provide proof of inequality.
[14]	In transparent service applications, an unlogged output is the collusion evidence.

party cannot generate a proof of collusion by themselves without causing framing issues. Kelkar et al. [14] considered transparent service applications, and a colluding party can present the output as evidence of collusion.

Incentive mechanism. In Table 2, we compare the mechanisms in the three works [10, 12, 14] according to 5 metrics previously mentioned [12] and 3 more metrics concerning the modeling choices. Since the second work [12] provides two mechanisms where the first mechanism extends the first work [10], we will single out the other mechanism and blend the first mechanism with the results in the prior work [10].

Before we explain the metrics, we briefly recall the notion of *access structures* and their *trackability*. Consider n parties. An access structure $\mathcal{A} \subseteq 2^{[n]}$ ($[n] = \{1, 2, \dots, n\}$) is a collection of groups of parties who can reconstruct the secret, and a minimal set is one who has no strict subset in $\mathcal{A}$. For $2 \leq k \leq n$, we will consider (monotone) k -homogeneous access structures: All minimal sets in it have size k . An access structure $\mathcal{A}$ is ω -trackable when any size- ω subset is contained in at most one minimal set, e.g., the threshold- k access structure $\{S \subseteq \{1, 2, \dots, n\} : |S| = k\}$. For example, an (n, k, ω) -design is ω -trackable: Every size- ω subset appears in a unique minimal set in this design.

We now explain the four less straightforward metrics below:

t_e -Effective: The mechanism induces the non-collusion outcome given up to t_e malicious parties.

t_f -Fair: The mechanism does not affect non-colluding parties given up to t_f malicious parties.

φ -Fairness hazard: With $(k-1)$ malicious parties, the maximum meaningful number to consider, the mechanism mislabels $\leq \varphi$ non-colluders as colluders.

t_r -Robust: The protocol-initiated reconstruction can be stalled only by $\geq t_r$ absentees. That is, reconstruction goes through with $< t_r$ absentees.

We desire the fairness hazard to be small and the other three metrics to be large. Note that $t_r \leq (n - k + 1)$ due to k -homogeneity. Naturally, it is only meaningful to consider $t_e, t_f < k$.

We next describe and compare the results in the three works as outlined in Table 2, with a focus on how the mechanisms handle strong coalitions, where parties may not act independently or are influenced by other external mechanisms.

Quantitative metrics. Since Kelkar et al. [14] do not explicitly consider malicious parties who behave arbitrarily including inducing self-harm, we focus on comparing the other two works on this metric. The first mechanism [10] originally applies to n -out-of- n or the singleton access structure, $\{N\}$, and can be extended to general access structures [12]. It is thus $(k-2)$ -effective, $(k-2)$ -fair, has $(n-k)$ -fairness hazard, and has optimal $(n-k+1)$ -robustness.

² We note that there are other evidence collection methods in privacy-targeted collusion if one has specialized constructions for the underlying primitive, such as accountable signatures [16], secret sharing with snitching [7] and traceable secret sharing [13].

Table 2: Comparison of three deterrence mechanisms (in chronological order). Some cells are colored to highlight the outperforming results. For fixed n, k , the function $f^{(n,k)}(\omega)$ is monotonically increasing in ω . “Repeatable” asks if the mechanism maintains the non-collusion equilibrium if the underlying primitive is repeated for unknown or infinite number of times. “Strong coalition” asks if the mechanism considers strong coalitions where rational parties no longer act independently [10] or act independently but are exposed to externality [14]. “Exit and collude” asks whether the mechanism considers adversarial exiting strategies where parties exit the system and collude to learn many secrets at once. “Blind collusion” asks if the mechanism considers collusion scenarios where parties may give incorrect inputs.

Mechanisms	<u>Malicious fault bounds</u>		Fairness hazard	Applicable access structures	Robustness
	Fairness	Effectiveness			
[10]	$k - 2$	$k - 2$	$n - k$	Any	$n - k + 1$
[12]	$k - 1$	$k - 1 - \omega$	0	ω -trackable	$f^{(n,k)}(\omega)$
[14]	0	0 ²	$n - k$	Any	$n - k + 1$

Mechanisms	Repeatable	Strong coalition	Exit and collude	Blind collusion
[10]	✓	✓	✓	✓
[12]	✓	-	-	- ¹
[14]	✗	✓ ³	✗	✗

¹The “strong coalition”, “exit and collude” and “blind collusion” scenarios are not explicitly modeled, but [12] builds on [10], and inherits the results of the prior work [10].

²Malicious parties who behave arbitrarily are not explicitly considered by Kelkar et al. [14].

³Gong et al. [10] considered strong, mutually trusted coalitions that do not report collusion inside the coalition, while Kelkar et al. [14] considered individual rational parties with “smart collusion” where colluding parties establish retaliatory contracts to penalize parties who betray the collusion group, which is also one approach to establish a strong coalition.

The second mechanism [12] improves on fairness metrics, and applies to ω -trackable access structures ($1 \leq \omega < k$). It is $(k - 1)$ -fair, $(k - 1 - \omega)$ -effective and has optimal 0 fairness hazard. Its robustness is characterized with an appropriate increasing function $f^{(n,k)}(\omega)$ that monotonically increases with ω for fixed n, k , indicating a trade-off between effectiveness and robustness when setting ω . Note that $f^{(n,k)}(\omega)$ is known only for a handful of parameters and remains an open problem for a other parameters (shown in Table 2 of [12]).

Repeated executions. Both works by Gong et al. [10, 12] consider the more natural settings of repeated executions of the underlying primitive. One open problem in the mechanism by Kelkar et al. [14] is to extend it to repeated runs especially when the underlying application is a service provided to clients. Applying the two methods in the other two works [10, 12], either through having a large number of parties to pick the actual computing parties from or setting a fixed term for each computing party’s participation, could help extend the feasibility results.

Strong coalitions. Gong et al. [10] considered strong coalitions, where members *never* reports collusion inside the coalition, and established the following results given a secret that is worth at most V to any individual computing party:

- Grand coalition, where all computing parties form a coalition and collude, is the equilibrium if the profits to a coalition of any size $s \in [2, n]$ is at least $s \cdot V$, i.e., increasing by at least V as each new party joins the coalition, such as a bus with enough space. In this case, a fair allocation of the coalition value to all members ensures that parties do not increase their utilities by deviating from forming a grand coalition.
- When the coalition value stays the same or even decreases as more parties join, e.g., a finite treasure, one can always design mechanisms that limit coalition size to 2. The maximum coalition size for general parameters was visualized here [11].

One open problem is to model “weaker” coalitions where each party may join multiple weaker coalitions that they can betray or negotiate with.

Kelkar et al. [14] considered retaliatory contracts that penalizes whistleblowing colluders. Parties are still analyzed individually but are now bound to the restrictions imposed by other mechanisms. Let the secret still be of worth V . The authors established the following results:

- If the retaliatory contract can impose unbounded amount of penalties, it is impossible to achieve the non-collusion outcome.
- If the retaliatory contract can only impose bounded amount of penalties, it is impossible to achieve the non-collusion outcome with a sufficiently large penalty.

The earlier work [10] can be trivially extended to accommodate known bounded externality B , e.g., bribes or side contracts, by increasing the penalty by slightly more than B . The later work [14] studies the relationship between the side contract and deterrence mechanism in more detail in the case where each party can singlehandedly produce proofs of collusion, which not necessarily holds for general evidence collection modules [10].

Overall, the two models to capture coalitions are complementary: one [10] analyzes colluders who are committed to coalition peers and handles coalitions when the secret decreases in value for individuals as more parties learn about it, while the other [14] digs into coalitions glued together via incentives and bounds the size of penalties in the retaliatory contract. In other words, [10] tackles stronger mutually trusted coalitions for exclusive secrets, and [14] tackles mutually distrusted individuals glued together by side contracts for secrets of more general nature.

Exit and collude. Gong et al. [10] considered adversarial exiting strategies where a party may exit the system and sell the received information, or collude with others to recover all possible secrets before exiting. In the first case, it is in general impossible to stop information selling without provable deletion. In the second case, one can set a privacy protection period so that the mechanism only protects recent secrets while requiring only finite payments.

Blind collusion. Gong et al. [10] additionally models blind collusion where computing parties may cheat during collusion to mislead other colluders. This means that in the collusion game, parties can choose whether to participate, whether to provide correct inputs, and whether to report collusion. Blind collusion requires a new solution concept since there is information asymmetry, i.e., before reporting, one may not know whether others have submitted correct inputs during collusion. Blind collusion applies to a large number of protocols secure against semi-honest parties where it is not computationally hard to forge inputs, e.g., secret sharing [21] (without verifiability) or basic multi-server PIR schemes [5].

The authors then designed a mechanism to induce non-collusion outcome in this case as well.

Trackable access structure (TAS). Gong et al. [12] introduced a new family of access structures called trackable access structure (TAS) that allow more precise location of colluders. They also

demonstrated constructions of optimal and near-optimal TAS and constructions of efficient secret sharing schemes on near-optimal TAS.

Yet, as noted in related works [10, 14], when the individual worth of the secret stays the same or increases as more parties learn about it, and parties can establish side contracts with unbounded penalties, precise blaming does not stop collusion.

Organization of contents. We will briefly summarize the preliminaries to understand and potentially extend the results in these works in Section 2. We then characterize the implications for input privacy protection in MPC in Section 3. We summarize open problems in Section 4.

2 Preliminary

2.1 Access structure

Definition 1 (Access structure). *Given a set $[n] = \{1, \dots, n\}$ of n parties, an access structure $\mathcal{A}$ on $[n]$ is a collection of subsets of $[n]$, $\mathcal{A} \subseteq 2^{[n]}$.*

A set in $\mathcal{A}$ is called an *access group* or an *authorized set*. We now define two key traits of interest, **robustness** and **trackability**.

Definition 2 (Robustness). *The robustness of an access structure $\mathcal{A}$, denoted as $r(\mathcal{A})$, is the minimum number of parties that need to be corrupted to make reconstruction impossible.*

Definition 3 (ω -trackability). *An access structure is ω -trackable if given any ω parties, there exists either none or a unique minimal set containing them.*

Equivalent definitions to partial Steiner systems, uniform subsets with restricted intersections and binary constant-weight code are also provided in [12]. For a k -homogeneous access structure defined on party set $[n]$ and with ω -trackability, we denote it as (n, k, ω) -TAS. Details on how to construct optimal and near-optimal TAS and how to construct efficient secret sharing schemes on this new access structure can be found in [12].

2.2 Game theory definitions

Game representation. In game theory, a normal-form game can be characterized by parties and their action space and utility functions. In our setting, let $N_m \subseteq [n]$ be the set of malicious parties where $|N_m| \leq k - 1$, $N_r = [n] - N_m$ be the set of rational parties ($|N_r| \geq (n - k + 1)$), $\times_{i \in N_r} A_i$ be the rational parties' joint action space with A_i being party i 's individual action space, and $\{u_i\}_{i \in N_r}$ be their utility functions. We can then represent the game as a tuple $(N_r, \times_{i \in N_r} A_i, \{u_i\}_{i \in N_r})$. Considering that parties' interactions are sequential in the proposed mechanisms, we additionally need to capture parties' knowledge of past actions and beliefs of others' future actions when describing the game.

Solution concepts. A *strategy* s_i of party i is a probability distribution $\mathcal{D}$ over action space A_i , with all mass concentrated at one action for a pure strategy. We denote i 's strategy space as $\mathcal{D}_i$. We denote malicious parties' joint strategy space as $\mathcal{D}_M$ (which is unknown). The utility function of i can then be described more explicitly as a function mapping all parties' strategies to real-valued utilities, $u_i : (\times_{j \in N_r} \mathcal{D}_j) \times \mathcal{D}_M \mapsto \mathbb{R}$. A *strategy profile* then records the strategies of all rational parties. Strategy profiles yielding certain desired properties are called *equilibria* or *solution concepts*.

For example, in a Nash equilibrium (NE), no party can increase its utility by unilaterally deviating from the equilibrium strategy.

In two prior designs [12, 14], the mechanisms are public knowledge, and parties act openly. This means that parties have complete information about the game structure and perfect information about historical moves. In this case, *Subgame Perfect Equilibrium* (SPE [20]) is a suitable solution concept. Informally, the equilibrium strategy profile specifies the NE strategies for each party for every subgame. Here, a subgame at a step (where any party needs to make a move) is the continuation of the complete game from this step. SPE is a refinement of NE as NE is susceptible to empty (non-credible) threats.

Definition 4 (SPE [20]). *A Nash equilibrium (NE) is a strategy profile $\mathbf{s}$ where no party increases utility by unilaterally deviating from $\mathbf{s}$. A subgame perfect equilibrium (SPE) is a strategy profile $\mathbf{s}$ that forms an NE for any subgame.*

In the first design [10] where blind collusion is additionally modelled, parties have unperfect information about others' past actions. As such, *Sequential Equilibrium* [15] is a suitable solution concept. Informally, in sequential equilibrium, players take turns to act, and they all hold beliefs over the probability distributions of others' future actions. They select the dominant strategies according to their beliefs, and their played actions confirm others' prior beliefs about their actions in retrospect.

To formalize the notion of sequential rationality, we first formalize the notion of a *belief*.

Definition 5. *For each information set I , a belief assessment b gives the conditional probability distribution $b(\cdot|I)$ over all decision nodes $v \in I$.*

Here, an *information set* for a player contains all decision nodes, i.e., steps where the player takes an action, indistinguishable by the player.

Definition 6 (Sequential equilibrium [15]). *A strategy profile and belief assessment pair $(\mathbf{s}, b)$ is a sequential equilibrium if $(\mathbf{s}, b)$ is sequentially rational and b is consistent with $\mathbf{s}$.*

Given $(\mathbf{s}, b)$, $\mathbf{s}$ is called *sequentially rational* if at each information set I , the player to take an action maximizes her expected payoff, given beliefs $b(\cdot|I)$ and that other players follow $\mathbf{s}$ in the continuation (remaining) game. Given a strategy profile $\mathbf{s}$, for any information set I on the path of play of $\mathbf{s}$, $b(\cdot|I)$ are *consistent* with $\mathbf{s}$ if and only if they are derived using Bayes' rule. For example, let player 1 be at an information set $I = \{x, y\}$. Suppose according to $\mathbf{s}$, x, y are respectively reached with probability 0.2, 0.3. Then a consistent belief for player 1 is that $b(x|I) = \frac{0.2}{0.2+0.3} = 0.4$ and $b(y|I) = 0.6$.

Fairness hazard. Given an equilibrium strategy profile $\mathbf{s}^*$, we now formally define "fairness". Denote the honest non-colluding strategy played by party i as s_i^h , the strategy profile of all other rational parties in equilibrium as $\mathbf{s}_{-i}^*$, and the arbitrary strategy profile of malicious parties as $\mathbf{s}_M$.

Definition 7 (Fairness). *For all $i \in N_r, \forall s_i \in \mathcal{D}_i, \forall \mathbf{s}_M \in \mathcal{D}_M$,*

$$u_i(s_i^h, \mathbf{s}_{-i}^*, \mathbf{s}_M) \geq u_i(s_i, \mathbf{s}_{-i}^*, \mathbf{s}_M)$$

This means that regardless of malicious parties' actions, the honest non-colluding strategy is the dominant strategy for rational parties in equilibrium. A collusion deterrence mechanism is fair if it ensures fairness for every non-colluding rational party in equilibrium. If a mechanism does not ensure fairness given $(k-1)$ malicious parties, we capture the risk of falsely implicating non-colluding parties with the *fairness hazard* notion.

Definition 8 (Fairness hazard). *Given $(k - 1)$ malicious parties, fairness hazard counts the number of parties among $(n - k + 1)$ rational parties whose utilities can be strictly increased by not playing the honest strategy.*

3 Implications for Input Privacy in MPC

Let us limit the discussion to honest majority MPC [3, 4, 18] and general outsourced MPC including PIR. We will focus on describing the case before output reconstruction, and we note that collusion after output reconstruction would require non-trivial efforts beyond applying prior works.

Before output reconstruction. In some applications, there is no reconstruction phase among the computing parties, such as PIR. In such cases, the three proposed schemes can be utilized to discourage collusion, although the recent scheme [14] requires an additional proper evidence collection module to function.

Specifically, let the input to be protected be x . If no other party has prior knowledge about the x , and x is not easily guessable², one can directly adopt the two schemes with the general evidence collection module [10, 12] while treating the underlying primitive as a black box.

However, if parties may possess prior knowledge about x , or x is easily guessable, one needs to update the underlying protocol to include dummy secrets picked uniformly at random [10, 12]. This inevitably blows up communication and computation costs.

4 Conclusion and Open Problems

The three collusion deterrence protocols utilize incentives to make privacy-targeted collusion over unobserved channels visible given enough rational parties. They optimize over different metrics for more practical collusion deterrence, e.g., effectiveness, fairness, coalition resistance, side contract resistance, etc. There are abundant open problems in the direction of deterring privacy-targeted collusion in MPC, and we present some open problems concerning deterrence mechanisms allowing general collusion methods (as the three compared works).

1. The first general open problem is to disincentivize collusion in generic MPC.
2. Another general open problem is to have more general mechanism implementations other than explicit rewards and penalties, such as reputation systems and credit networks. This would expand the usability and applicability of such mechanisms.
3. It would be useful to devise tighter evidence collection modules. [14] abstracts away generation of evidence of privacy-targeted collusion, and [10, 12] requires proving triviality of functions during evidence collection. There is a gap between provably trivial and provably non-trivial functions (up to the same triviality parameter), so closing the gap or deriving more suitable evidence collection module without compromising the generality of collusion methods would improve the practicality of collusion deterrence mechanisms.
4. It would be useful to analyze strong coalitions more rigorously, e.g., allow overlapping of strong coalitions. Prior work [10] only considered the case when strong coalitions are non-overlapping, and each party is loyal to at most one strong coalition.
5. In the TAS-based mechanism [12], there are many open problems related to TAS, such as counting the robustness number of TAS with general parameters, constructing optimal TAS, and constructing efficient secret sharing or MPC schemes on TAS.

² Easiness is parameterized into the mechanism and computed with min-entropy.

6. In the mechanism based on “smart collusion” [14] where coalitions are glued together with side contracts, one can further consider different natures of secrets and expand the feasibility results. Additionally, one may consider repeated executions, blind collusion and adversarial exiting strategies.

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